

Every finite-dimensional nowhere-differentiable subspace extends to a closed continuum-dimensional one

Candidate full solution for arXiv:2306.01561

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Status. Candidate full proof, likely valid, requiring human review. The result affirmatively answers the stated $(1, \mathfrak{c})$ -spaceability question and proves the stronger $(n, \mathfrak{c})$ -spaceability statement for every finite positive integer n .

1 Source question

Let $C[0, 1]$ be the real Banach space with the supremum norm, and let

$$\mathcal{ND}[0, 1] = \{f \in C[0, 1] : f \text{ is nowhere differentiable}\}.$$

Araújo, Barbosa, Raposo Jr., and Ribeiro state on PDF page 4 of [1] that $\mathcal{ND}[0, 1]$ is known to be spaceable, but ask whether it is specifically $(1, \mathfrak{c})$ -spaceable. In other words: does every prescribed one-dimensional subspace contained in $\mathcal{ND}[0, 1] \cup \{0\}$ lie in a closed $\mathfrak{c}$ -dimensional subspace contained in $\mathcal{ND}[0, 1] \cup \{0\}$?

2. SPACEABILITY IN COMPLEMENT OF UNIONS

In this section, we delve into a comprehensive analysis of spaceability in the complements of unions of subspaces, aiming to answer a fundamental question: *what conditions are necessary and sufficient to ensure that the complement of a union, not necessarily countable, is spaceable?*

So far, we do not have a complete set of conditions that characterize spaceability in this complex context. However, works like that of Kitson and Timoney in [26] offer intriguing insights, highlighting that under certain conditions, the complement of a countable union of subspaces is spaceable. Furthermore, we aim to understand more about the relationship between the notions of *spaceability* and (α, β) -*spaceability*. For example, the set of functions nowhere differentiable in $C[0, 1]$, denoted by $\mathcal{ND}[0, 1]$, is one of those sets that can be described as the complement of a union of dense subspaces. According to the literature (see [1, 24]), it is spaceable, but it has not yet been determined whether it is specifically $(1, \mathfrak{c})$ -spaceable or not.

Figure 1: The exact open-question passage in [1, p. 4], rendered and cropped from the local source-paper PDF.

We use one quantitative theorem of Hencl. In [2, Proposition 2], Hencl proves that for every continuous increasing function $\varphi : [0, \infty) \rightarrow [0, \infty)$ with $\varphi(0) = 0$, there are a Cantor set $K \subset [0, 1]$ and a linear isometry

$$T : C(K) \longrightarrow C[0, 1]$$

such that the following holds. For every $0 \neq u \in C(K)$, $z \in (0, 1)$, and $m \in \mathbb{N}$, the set

$$D_z(u, m) = \left\{ y \in [0, 1] \setminus \{z\} : \frac{|Tu(y) - Tu(z)|}{\varphi(|y - z|)} > m \right\} \quad (1)$$

has symmetric upper density one at z . At 0 and 1 it has, respectively, right and left upper density one. Hencl also obtains $Tu = u$ on K , though that feature is not needed here.

2 Proof intuition

A prescribed finite-dimensional subspace $E \subset C[0, 1]$ has a common modulus of continuity: compactness of its unit ball makes the oscillations of all its unit vectors uniformly small at small scales. Choose Hencl's embedding to oscillate much more rapidly than that common modulus. Then no fixed element of E can cancel the small-scale oscillations of any nonzero Hencl function.

Consequently the Hencl range is disjoint from E except at zero, and every nonzero vector in their algebraic sum is still nowhere differentiable. The Hencl range is closed because the embedding is isometric, and adding the finite-dimensional space E preserves closedness. This produces the required closed extension without altering the prescribed directions.

3 A common modulus for a finite-dimensional subspace

Let E be a nonzero finite-dimensional subspace of $C[0, 1]$. For $0 \leq t \leq 1$, define

$$\Omega_E(t) = \sup\{|e(x) - e(y)| : e \in E, \|e\|_\infty \leq 1, x, y \in [0, 1], |x - y| \leq t\}. \quad (2)$$

Lemma 1. *The function Ω_E is increasing, subadditive, and continuous, with $\Omega_E(0) = 0$ and $\Omega_E(t) \rightarrow 0$ as $t \downarrow 0$. If E contains a nonconstant function, then for some $c > 0$ and all sufficiently small $t > 0$,*

$$\Omega_E(t) \geq ct.$$

Proof. The closed unit ball of E is compact in the supremum norm. A finite ε -net, together with uniform continuity of its finitely many members, shows that this ball is uniformly equicontinuous. Hence $\Omega_E(t) \rightarrow 0$ as $t \downarrow 0$.

Monotonicity is immediate. To prove subadditivity, suppose $|x - y| \leq s + t$. Choose a point between x and y so that the two resulting distances are at most s and t . Applying the triangle inequality and then taking the supremum in (2) gives $\Omega_E(s + t) \leq \Omega_E(s) + \Omega_E(t)$ when $s + t \leq 1$. Thus $|\Omega_E(s) - \Omega_E(t)| \leq \Omega_E(|s - t|)$, which, with the limit at zero, proves continuity.

Finally choose $e_0 \in E$ with $\|e_0\|_\infty = 1$ and points $a < b$ for which $\delta = |e_0(b) - e_0(a)| > 0$. Partition $[a, b]$ into $N = \lceil (b - a)/t \rceil$ intervals of length at most t . The triangle inequality yields

$$\delta \leq N\Omega_E(t) \leq \left(\frac{b - a}{t} + 1 \right) \Omega_E(t),$$

and therefore $\Omega_E(t) \geq \delta t / (b - a + t)$. This gives the claimed linear lower bound near zero. $\square$

4 The finite-extension theorem

Theorem 1. *For every $n \in \mathbb{N}$, the set $\mathcal{ND}[0, 1]$ is $(n, \mathfrak{c})$ -spaceable in $C[0, 1]$. Explicitly, if $E \subset C[0, 1]$ is n -dimensional and*

$$E \setminus \{0\} \subset \mathcal{ND}[0, 1],$$

then there is a closed subspace $M \subset C[0, 1]$ such that

$$E \subset M, \quad \dim M = \mathfrak{c}, \quad M \setminus \{0\} \subset \mathcal{ND}[0, 1].$$

The same proof rules out finite one-sided derivatives at the endpoints.

Proof. Fix such an E . Since every nonzero member of E is nowhere differentiable, E contains a nonconstant function. Extend Ω_E constantly from $[0, 1]$ to $[0, \infty)$ and set

$$\varphi(t) = \sqrt{\Omega_E(t)}.$$

By Lemma 1, φ is continuous and increasing, with $\varphi(0) = 0$. Moreover,

$$\frac{\Omega_E(t)}{\varphi(t)} = \sqrt{\Omega_E(t)} \rightarrow 0, \quad \frac{\varphi(t)}{t} \rightarrow \infty \quad (t \downarrow 0). \quad (3)$$

The second assertion follows from $\Omega_E(t) \geq ct$ near zero.

Apply Hencl's proposition to this φ , and denote the resulting isometry by $T : C(K) \rightarrow C[0, 1]$. Put $Y = T(C(K))$.

We first prove the key perturbation statement. Fix $0 \neq u \in C(K)$, $e \in E$, and $z \in [0, 1]$. Because $D_z(u, 1)$ in (1) has upper density one at z (with the stated one-sided interpretation at an endpoint), it contains a sequence $y_j \neq z$ converging to z from the allowed side. By the definition of Ω_E ,

$$|e(y_j) - e(z)| \leq \|e\|_\infty \Omega_E(|y_j - z|). \quad (4)$$

The first limit in (3) shows that, for all sufficiently large j , the right side of (4) is at most $\varphi(|y_j - z|)/2$. On the other hand, membership in $D_z(u, 1)$ gives $|Tu(y_j) - Tu(z)| > \varphi(|y_j - z|)$. Hence

$$\begin{aligned} |(Tu + e)(y_j) - (Tu + e)(z)| &\geq |Tu(y_j) - Tu(z)| - |e(y_j) - e(z)| \\ &> \frac{1}{2} \varphi(|y_j - z|). \end{aligned}$$

After division by $|y_j - z|$, the second limit in (3) shows that these difference quotients are unbounded. Thus $Tu + e$ has no finite derivative at z . Since z was arbitrary,

$$0 \neq u \in C(K), \quad e \in E \quad \implies \quad Tu + e \in \mathcal{ND}[0, 1]. \quad (5)$$

In particular $Y \cap E = \{0\}$: if $Tu = e \in E$ with $u \neq 0$, then (5) applied to $Tu - e = 0$ would assert that the zero function is nowhere differentiable, a contradiction. Now define

$$M = Y + E = Y \oplus E.$$

The range Y is closed because T is an isometry. Also $Y + E$ is closed: if $q : C[0, 1] \rightarrow C[0, 1]/Y$ is the quotient map, then $q(E)$ is finite-dimensional and hence closed, while $Y + E = q^{-1}(q(E))$.

Every nonzero element of M has the unique form $Tu + e$. If $u \neq 0$, it belongs to $\mathcal{ND}[0, 1]$ by (5); if $u = 0$, then $e \neq 0$ and the hypothesis on E applies. Finally Y is isometric to $C(K)$, whose Hamel dimension is $\mathfrak{c}$, so $\dim M = \mathfrak{c}$. This proves all assertions. $\square$

Corollary 1. *The $(1, \mathfrak{c})$ -spaceability question in [1] has an affirmative answer.*

5 Verification, scope, and novelty

The proof is noncomputational. Its load-bearing points are: compactness of the unit ball of a finite-dimensional subspace; the two scale comparisons in (3); the upper-density quantifier in Hencl’s Proposition 2; and closedness of a finite-dimensional extension of a closed subspace. The accompanying `verification.md` records adversarial checks of each.

The theorem concerns real-valued continuous functions, exactly as in the source passage. It proves the positive finite-dimensional part. Ribeiro’s 2024 thesis independently proves that $\mathcal{ND}[0, 1]$ is not (α, β) -spaceable when α is infinite [3, Corollary 2.2]; together, the two statements give the finite-versus-infinite classification for prescribed starting dimension.

Bounded novelty check. On 11 August 2026, the local run indexes were searched by arXiv id, title, and the terms “nowhere differentiable”, “ $(1, c)$ -spaceable”, “pointwise spaceability”, “Hencl”, and “finite-dimensional extension”. External exact and close searches used the same phrases and checked the source/publisher record, Hencl’s theorem, Ribeiro’s 2024 thesis, and work indexed through August 2026. The thesis’s pointwise-spaceability result is for a different set of unbounded integrable functions; for $\mathcal{ND}[0, 1]$ it supplies only the negative infinite-dimensional obstruction. The July 2026 preprint of Lima, Santos, and Ribeiro proves that every prescribed finite-dimensional subspace extends to a countably infinite-dimensional *algebraic* subspace [4]. It does not claim a closed extension or continuum dimension, and it does not invoke Hencl’s quantitative embedding. No prior statement of the theorem above or of this perturbation argument was found. Novelty confidence is moderate rather than definitive; specialist review of non-indexed theses and citing literature is recommended.

References

- [1] G. Araújo, A. Barbosa, A. Baganha Raposo Jr., and G. Ribeiro, *Complements of unions: insights on spaceability and applications*, *Mathematika* **71** (2025), e70006; [arXiv:2306.01561](#).
- [2] S. Hencl, *Isometrical embeddings of separable Banach spaces into the set of nowhere approximately differentiable and nowhere Hölder functions*, *Proc. Amer. Math. Soc.* **128** (2000), 3505–3511, [doi:10.1090/S0002-9939-00-05595-7](#).
- [3] G. dos Santos Ribeiro, *A search for linearity in the universe of topological vector spaces*, Ph.D. thesis, Universidade Federal da Paraíba, 2024, [institutional repository record](#).
- [4] F. Lima, M. Santos, and G. Ribeiro, *On the $(n, \aleph_0)$ -lineability of nowhere differentiable functions*, [arXiv:2607.15749](#) (2026).